

EXAM 7 – SPRING 2012

8. (3.75 points)

Given the following loss development triangle and selections:

Accident Year	<u>Incremental Incurred Losses (\$000)</u>				
	0-12 Months	12-24 Months	24-36 Months	36-48 Months	48-60 Months
2007	5,800	1,754	875	310	54
2008	6,000	1,675	780	285	
2009	5,700	1,850	888		
2010	5,850	1,500			
2011	6,300				

<u>Selected Age-to-Age Link Ratios</u>				
12-24 Months	24-36 Months	36-48 Months	48-60 Months	60 Months to Ultimate
1.290	1.112	1.035	1.006	1.000

- Incremental losses, C_{ij} , follow an over-dispersed Poisson distribution with mean $x_i y_j$ and variance $\phi x_i y_j$.
- The variable x_i represents the expected ultimate losses for accident year i .
- The variable y_j represents the proportion of ultimate losses that emerge in development year j .
- The prior distribution for x_i is gamma with mean α_i/β_i and variance $\alpha_i/(\beta_i)^2$.
- The dispersion parameter, ϕ , for the over-dispersed Poisson distribution is 8.429.
- The accident year 2010 estimates for α and β are 100 and 0.01143 respectively.

The mean of C_{ij} for this Bayesian model is:

$$Z_{ij} \times (\lambda_j - 1) \times D_{i,j-1} + (1 - Z_{ij}) \times (\lambda_j - 1) \times x_i \times \left(\frac{1}{\lambda_j} \right)$$

where:

$$Z_{ij} = \frac{\sum_{k=1}^{i-1} y_k}{\beta_i \times \phi + \sum_{k=1}^{i-1} y_k}$$

- λ_j is the incremental chain ladder loss development factor for development year j .
- D_{ij} is the cumulative losses for accident year i as of development year j .

QUESTION 8 CONTINUED ON NEXT PAGE

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(8 continued)

a. (3 points)

Use this model to calculate the incremental losses for accident year 2010 expected to emerge between 24 and 36 months of development.

b. (0.75 point)

Explain how this model can be interpreted as a trade-off between the standard chain ladder method and the Bornhuetter-Ferguson method.

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Question 8 Sample Solution

Solution 1

a)

$$\text{expected } c_{ij} = \frac{\alpha}{\beta} = \frac{100}{.01143} = 8748.91 \quad 24-36 \text{ for AY10}$$

$$\begin{aligned} \text{LDF}_{12-\text{ult}} &= 1.29 \cdot 1.112 \cdot 1.035 \cdot 1.006 \\ &= 1.4936 \end{aligned}$$

$$\text{LDF}_{24-\text{ult}} = 1.1578$$

$$1/\text{LDF} = D_k = \frac{0-12}{.66952} \quad \frac{12-24}{.1941814}$$

$$Z = (.66952 + .1941814) / (.01143 \cdot 8.429 + .66952 + .1941814) = .899647$$

$$\begin{aligned} C_{ij} &= .899647 \times (1.112-1) \cdot (5850+1500) + (1-.899647)(1.112-1) \cdot (8748.91/1.112) \\ &= \underline{\underline{829.02}} \end{aligned}$$

Solution 2

a)

	<u>12-Ult</u>	<u>24-Ult</u>	<u>36-Ult</u>	<u>48-ult</u>	<u>60-Ult</u>
% Rptd.	1.4936 66.95%	1.1578 86.37%	1.04121 96.04%	1.006 99.4%	1.000 100%

$$Z_{24-36} = \frac{86.37\%}{(8.429)(0.01143) + 86.37\%} = 0.8996$$

$$\lambda_{24-36} = 1.112$$

$$D_{2010,24} = 5850 + 1500 = 7350$$

$$x_{2010} = \text{mean of Gamma} = \frac{\alpha}{\beta} = \frac{100}{0.01143} = 8749$$

$$\begin{aligned} \text{Incremental losses AY 2010 btw 24 and 36 months} &= \\ 7350(1.112 - 1) \times 0.8996 + 8749(96.04\% - 86.37\%) \times (1-.8996) &= \\ = 740.55 + 84.941 &= \\ = 825.49 \end{aligned}$$

Solution 3

a) CL Estimate:

$$\begin{aligned}\text{Expected incremental 2012 IBNR losses @ 24-36m} \\ &= (1,500 + 5,850)(1.112 - 1) \\ &= 823,200\end{aligned}$$

BF Method Estimate:

$$\text{Expected losses} = \alpha/\beta = 100/.01143 = 8,749$$

$$\begin{aligned}\text{IBNR Increment Estimate} &= \text{Exp losses} \times (D_{36-48} - D_{24-36}) \rightarrow \text{Need } D_{24-36} \\ D_{24-36} &= (1/1.041) - (1/1.158) = 9.71\%\end{aligned}$$

$$\text{Cum LDF:} \quad \overset{24:36}{1.158} = (1.041)(1.112) \quad \overset{36:48}{1.041} = (1.035)(1.006)$$

$$\Rightarrow \text{IBNR BF estimate} = 8,749(.0971) = 849,151$$

$$Z = \frac{\frac{1}{1.158}}{.01143(8.429) + \frac{1}{1.158}} = \frac{D_{k-1}}{\beta\phi + D_{k-1}} = .8996$$

$$\begin{aligned}\text{2010 Inc Losses Estimate} &= Z \times \text{CL estimate} + (1 - Z) \text{ BF Estimate} \\ &= .8996(823,200) + (1-.8996) (849,151) \\ &= \$825,805\end{aligned}$$

Solution 4

a) Cum losses AY 2010 @ 24 months = 5850+1500 =

$$\begin{array}{ccccccc}\text{Cum LDF} & 12\text{-ult} & 24\text{ult} & 36\text{ult} & 48\text{ult} & & \\ & & 1.1578 & & & & \end{array}$$

$$Z_{2010,24} = \frac{\frac{1}{1.1578}}{0.01143 \times 8.429 + \frac{1}{1.1578}} = 0.8996$$

$$C_{2010,36} = \underbrace{Z \times (1.1578 - 1) \times 7350}_{1043.38} + \underbrace{(1 - Z)(1.1578 - 1) \times \frac{100}{0.01143} \times \frac{1}{1.1578}}_{119.7184}$$

$$\boxed{= 1163}$$

Solution1

- b) Notice that the equation is $Z \cdot (\lambda - 1) \cdot D + (1 - Z) (\lambda - 1) \cdot x \cdot (1/\lambda)$
- $(\lambda - 1) \times D$ is a chain ladder estimate of incremental loss.
 - $(\lambda - 1) \cdot x/\lambda$ is an expected emergence of our prior distribution.

Then both are credibility weight by Z , a factor that considers losses reported to date giving more weight to CL as more loss are reported. BF is similar in that it weights CL estimate with an expected ultimate based on Z of % loss reported.

Solution 2

- b) The credibility weighting uses the chain ladder estimate $(Z(\lambda_j - 1) D_{ij-1})$ and the Bornhuetter Ferguson estimate $((1 - Z)(\lambda_j - 1) x_i \times 1/\lambda_j)$ weighted by Z and $(1 - Z)$ respectively.

Examiner Comment

The model solution is based on the original paper interpretation of calculating $E[C]$ rather than the formula in the exam. There were a couple differences between the formula on the exam and the formula in the paper. Because of the potential confusion, a number of model solutions were given full credit.

The formula for the mean of C_{ij} for the Bayesian model is slightly different from the exam problem. Instead of $1/\lambda_j$, the correct formula is $1/(\lambda_j \lambda_{j+1} \dots \lambda_n)$. Candidates who used this formula instead will receive full credit as well. Additionally, candidates who used alternative formulas to calculate the BF estimate were given full credit as long as the method was accurate.

Additionally, the top of the summation term when calculating Z should have been $j-1(j-1=2)$ rather than $i-1(i-1=3)$. Most candidates ignored this difference but both answers were accepted.

Some of the common errors where candidates lost points include:

- Using the incremental LDF, or percent reported between 24 and 36 months, to calculate y
- Calculating x using the chain-ladder method rather than using α and β .
- Incorrect calculation of λ (e.g. using 36 to ultimate rather than 24 to ultimate)

The majority of candidates had little problem with the b. part. To receive full credit, candidates must discuss how the formula is a credibility weighted average of the chain ladder or BF method and must identify which part of the formula was C-L and which part of the formulas was BF.

Due to confusion in part a., some candidates discussed how the second half of the formula was “like a BF” but not exactly. These responses received full credit as well.

The most common errors were not mentioning “credibility” or “Z,” and confusing the chain-ladder and B-F terms.